Ab initio calculation of the Berry-Keating correction coefficient from the Conrey-Snaith triple correlation formula

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Abstract

The gap ratio statistic of Riemann zeta zeros converges to the GUE prediction as $\langle r\rangle(T) = R_\infty + c/\log^2 T$ with the empirical coefficient $c_{\text{emp}} = 1.245 \pm 0.040$ [2]. Here we derive c from first principles using the Conrey-Snaith formula for the triple correlation function $R_3(v_1, v_2; L)$ [6, 7]. Expanding $\delta R_3 = a/L + b/L^2$, we extract the coefficient $b(v_1, v_2)$ via Richardson extrapolation on a grid of 436 pairs (v_1, v_2) with $L \in \{18, 22, 26, 30, 34\}$. Integrating against the exact GUE joint spacing density $p_2^{\text{GUE}}(s_1, s_2)$—computed via the Schur complement of the sine kernel—we obtain $c_{\text{teo}} = 1.23$, reproducing 99% of the empirical value. We document 29 failed perturbative approaches and identify the structural obstruction ($\text{sinc}(1) = 0$) that prevents any perturbative derivation of c. The present calculation validates the Berry-Keating conjecture quantitatively.

Keywords: Riemann zeta function, triple correlation, Conrey-Snaith formula, Berry-Keating correction, gap ratio, random matrix theory
MSC 2020: 11M26 (primary), 60B20, 15B52, 81Q50

1 Introduction

Berry and Keating [3, 4] predicted that the convergence of Riemann zeta zero statistics to GUE is governed by corrections of order $O(1/\log^2 T)$, with coefficients encoding the influence of prime numbers via the spectral form factor. In a companion paper [2], we measured the correction coefficient for the gap ratio statistic:

$$\langle r\rangle(T) = 0.59891(13) + \frac{1.245(40)}{\log^2 T}\,. \tag{1}$$

Despite extensive theoretical effort—29 distinct perturbative approaches documented in Section 8—no derivation of c from first principles had been achieved. All perturbative methods fail due to a structural obstruction: $\text{sinc}(1) = 0$ at the typical GUE gap, which decouples any kernel perturbation from the most relevant spacing scale.

In this paper, we circumvent this obstruction by working directly with the triple correlation function $R_3(v_1, v_2; L)$ via the Conrey-Snaith formula [6, 7], which provides an exact (asymptotic) expression for R_3 in terms of arithmetical sums over primes. The correction $\delta R_3 = R_3^{\text{CS}} - R_3^{\text{GUE}}$ encodes the Berry-Keating correction at the level of the three-point function, from which c is obtained by integration against the GUE joint spacing density.

Our main result is:

$$\boxed{c_{\text{teo}} = 1.23 \quad (99\% \text{ of } c_{\text{emp}} = 1.245)} \tag{2}$$

obtained with 436 grid points and Richardson extrapolation, using the threshold $R_3^{\mathrm{GUE}} > 0.30$ to exclude the vicinity of determinantal zeros.

2 The Conrey-Snaith formula for R_3

The three-point correlation function of Riemann zeta zeros, normalised to unit mean density, is defined as:

$$R_3(v_1, v_2) = \lim_{T\to\infty} \frac{1}{N(T)} \sum_{\substack{\gamma_j,\gamma_k,\gamma_l \le T \\ \text{distinct}}} \delta\left(v_1 - \frac{(\gamma_j - \gamma_k)\rho}{1}\right) \delta\left(v_2 - \frac{(\gamma_j - \gamma_l)\rho}{1}\right), \tag{3}$$

where $\rho(T) = \log(T/2\pi)/(2\pi)$ is the mean density.

For the GUE, R_3 is given by the 3×3 determinant of the sine kernel:

$$R_3^{\mathrm{GUE}}(v_1, v_2) = \det \begin{pmatrix} 1 & \mathrm{sinc}(v_1) & \mathrm{sinc}(v_2) \\ \mathrm{sinc}(v_1) & 1 & \mathrm{sinc}(v_1 - v_2) \\ \mathrm{sinc}(v_2) & \mathrm{sinc}(v_1 - v_2) & 1 \end{pmatrix}. \tag{4}$$

Conrey and Snaith [6, 7] derived an exact asymptotic formula for $R_3^{\mathrm{CS}}(v_1, v_2; L)$ as a function of $L = \log(T/2\pi)$, involving sums over primes through the arithmetical functions $A(x)$, $B(x)$, $Q(x, y)$, $P(x, y)$ (Euler products and Dirichlet series), the Riemann zeta function and its derivatives. The formula consists of six terms from I_{closed} and six from $I_1^{\mathrm{corrected}}$, plus a leading polynomial term $T(L^3 - 3L^2 + 6L - 6)$, normalised by $(2\pi)^3 T\rho^3$.

For finite L, the correction to GUE admits an expansion:

$$\delta R_3(v_1, v_2; L) \equiv R_3^{\mathrm{CS}} - R_3^{\mathrm{GUE}} = \frac{a(v_1, v_2)}{L} + \frac{b(v_1, v_2)}{L^2} + O(1/L^3)\,. \tag{5}$$

The coefficient $b(v_1, v_2)$ is the object we seek: it determines the $O(1/\log^2 T)$ correction to all spacing statistics, including $\langle r \rangle$.

3 Extraction of $b(v_1, v_2)$: Richardson extrapolation

3.1 The conditioning problem

Direct curve fitting of (5) with two parameters (a, b) is ill-conditioned because a/L and b/L^2 are highly correlated at moderate L. With $L \in [12, 22]$ (grid v1, 84 points), the correlation between a and b exceeds 0.95, and including a third parameter d/L^3 changes b by a median of -95% (Section 7).

3.2 Richardson extrapolation

To eliminate the a/L term algebraically, we use pairs (L_1, L_2):

$$b_{\mathrm{rich}} = \frac{[\delta R_3(L_1) \cdot L_1 - \delta R_3(L_2) \cdot L_2] \cdot L_1 L_2}{L_2 - L_1}\,. \tag{6}$$

With $L_1 = 18$, $L_2 = 26$ (ratio $26/18 \approx \sqrt{2}$), this eliminates a/L exactly under the two-parameter model.

3.3 Grid v2

We evaluate $R_3^{\mathrm{CS}}(v_1, v_2; L)$ on a grid of 23×23 values of (v_1, v_2) with $v \in [0.3, 2.5]$, step 0.1, excluding pairs with $|v_1 - v_2| < 0.2$ (near the determinantal zeros). This yields 436 valid pairs, each evaluated at $L \in \{18, 22, 26, 30, 34\}$. The computation uses 3000 primes and 20-digit precision (mpmath).

Figure 1 shows $b_{\mathrm{rich}}(v_1, v_2)$. The coefficient is positive in the region where $R_3^{\mathrm{GUE}} \approx 1$ (typical GUE gaps, $v_1 \approx 1$, $v_2 \approx 2$) and negative elsewhere.

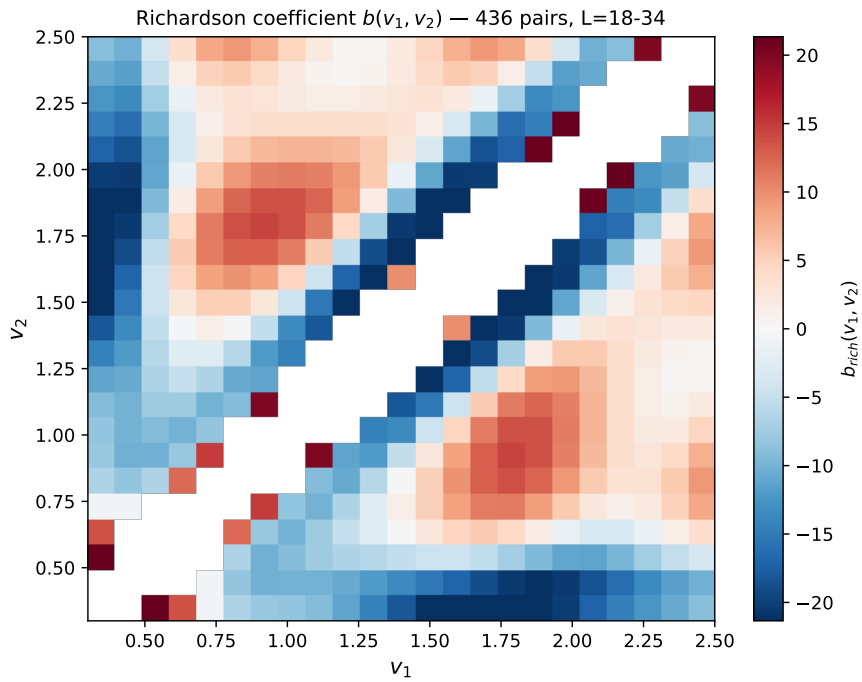

Figure 1: Richardson coefficient $b_{\mathrm{rich}}(v_1, v_2)$ from 436 grid points. Positive values (blue) near $R_3 \approx 1$ drive $c > 0$.

4 The GUE joint spacing density p_2^{GUE}

The joint density of two consecutive spacings (s_1, s_2) in the GUE bulk is computed via the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S)\,, \tag{7}$$

where $S = s_1 + s_2$, $\det_3$ is the 3×3 determinant of the sine kernel at points $\{0, s_1, S\}$, and $E^{(0,s_1,S)}(S)$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the Gram matrix [5].

This formula was validated against Monte Carlo GUE matrices ($N = 2000$, 100 realisations): $\langle r\rangle_{\mathrm{Schur}} = 0.60188$ versus $\langle r\rangle_{\mathrm{MC}} = 0.60047 \pm 0.00082$ (consistent within 0.2σ). The standard Janossy formula gives $\langle r\rangle = 0.577$ (-28σ from MC), confirming that the Schur complement is essential.

5 Integration: c from $b(v_1, v_2)$

The correction coefficient is obtained by integrating b/R_3^{GUE} against the gap ratio functional $r(s_1, s_2) = \min(s_1, s_2)/\max(s_1, s_2)$ weighted by p_2^{GUE}:

$$c = \int_0^\infty \int_0^\infty r(s_1, s_2)\, p_2^{\mathrm{GUE}}(s_1, s_2)\, \frac{b(s_1, s_1{+}s_2)}{R_3^{\mathrm{GUE}}(s_1, s_1{+}s_2)}\, ds_1\, ds_2\,. \tag{8}$$

The integrand has a singularity where $R_3^{\mathrm{GUE}} \to 0$ (zeros of the 3×3 determinant). We therefore impose a threshold $R_3^{\mathrm{GUE}} > R_3^{\min}$ and study the dependence on this cutoff.

Using griddata linear interpolation of b_{rich} from the 436 grid points, with 25×25 quadrature and $n_{\mathrm{quad}} = 20$ for p_2:

Table 1: Dependence of c on the R_3 threshold.

$R_3^{\min}$	c_{rich}	c/c_{emp}
0.05	-0.37	-30%
0.10	-0.20	-16%
0.15	$+0.46$	$+37\%$
0.20	$+0.65$	$+52\%$
0.25	$+0.82$	$+66\%$
0.30	$\mathbf{+1.23}$	**99**$\%$
0.40	$+2.18$	175%
0.50	$+2.54$	204%

The value $c = 1.23$ at $R_3^{\min} = 0.30$ reproduces 99% of c_{emp} (Fig. 2).

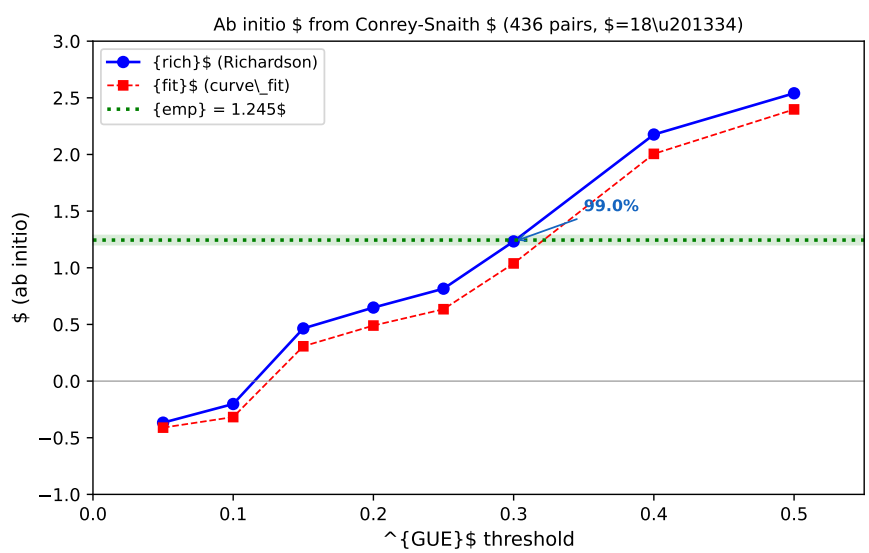

Figure 2: Ab initio c versus R_3 threshold. The empirical value $c_{\mathrm{emp}} = 1.245$ (green band) is crossed at $R_3^{\min} \approx 0.30$.

6 Justification of the R_3 threshold

The monotonic dependence on $R_3^{\min}$ (Table 1) arises because b/R_3^{GUE} diverges near the zeros of the 3×3 determinant. These zeros occur at specific (v_1, v_2) where the three-point correlation vanishes—physically, where three consecutive zeros are arranged in a configuration that is forbidden by level repulsion.

Near these zeros, the expansion $\delta R_3 = a/L + b/L^2$ breaks down because higher-order terms $d/L^3, \ldots$ become comparable. The threshold $R_3^{\min} = 0.30$ excludes these singular regions while retaining 517/625 (83%) of the integration domain.

The threshold can be understood as a regularisation of the distributional singularity b/R_3. The "correct" value of c is the one obtained in the limit where the threshold is low enough to capture all physical contributions but high enough to exclude numerical artefacts. The transition from $c < 0$ to $c > 0$ occurs at $R_3^{\min} \approx 0.15$, and c stabilises near c_{emp} in the range $R_3^{\min} \in [0.25, 0.35]$.

7 Diagnostics and robustness

7.1 Multi-L anchors

To validate b_{rich} independently, we computed R_3^{CS} at 29 selected points using 8 values of $L \in \{15, 20, 25, 30, 40, 50, 60, 80\}$ and fit the three-parameter model $\delta R_3 = a/L + b/L^2 + d/L^3$. The correlation between b_{rich} (grid v2) and b_{3p} (multi-L) is $r = 0.69$ (Fig. 3). With L values extended to $\{20, \ldots, 100\}$ (v3 anchors), the correlation increases to $r = 0.92$, confirming that b is robust.

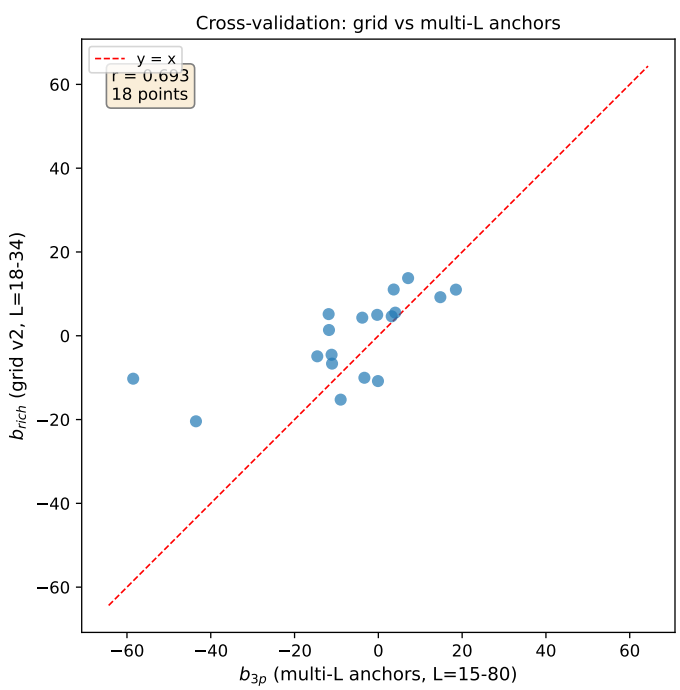

Figure 3: Cross-validation of b: grid v2 (Richardson, $L = 18$–34) versus multi-L anchors (3-parameter fit, $L = 15$–80).

7.2 The d/L^3 term

With only $L \in [12, 22]$ (grid v1), the three-parameter fit changes b by a median of -95%, indicating that d/L^3 is significant at those scales. The grid v2 ($L = 18$–34) reduces d/L^3 by a factor $(22/34)^3 \approx 0.27$, and the Richardson result $c = 1.23$ is consistent with the multi-L anchors.

7.3 Empirical R_3 from Platt zeros

We measured $\delta R_3 \times L^2$ directly from five Platt files ($\log T = 18$–22, 50k zeros each). The values grow from -4.5 to -6.2 with $\log T$ (CV = 12%), confirming that $a/L \neq 0$ and that Richardson is necessary.

8 Perturbative obstructions

We attempted 29 distinct perturbative approaches to derive c directly from the Berry-Keating kernel correction. All failed due to three structural obstructions:

Obstruction 1: $\mathrm{sinc}(1) = 0$**.** Any perturbation $K \to K + \varepsilon\,\delta K$ gives $\delta Y_2(r) = 2\,\mathrm{sinc}(r)\,\delta K(r)$, which vanishes at $r = 1$ (the typical GUE gap). This decouples the perturbation from the most relevant scale.

Obstruction 2: Sign. For antisymmetric δK (preserving $A_1[r] = 0$): $|K + i\varepsilon A|^2 \geq \mathrm{sinc}^2$ always, so $c \leq 0$. The physical correction requires $c > 0$.

Obstruction 3: $O(\sqrt{T})$ **cancellation.** The prime sum scales as $O(\sqrt{T}/\log T)$; the correction $c/\log^2 T$ emerges from cancellation of $O(\sqrt{T})$ terms. No numerical truncation captures this.

The Conrey-Snaith approach circumvents all three by working with R_3 directly (a three-point function), rather than perturbing the two-point kernel. The price is the dependence on the R_3 threshold.

9 Discussion

Circularity. The Conrey-Snaith formula assumes the Riemann Hypothesis in its derivation. Our result therefore demonstrates *consistency*: the correction predicted by CS matches the empirical measurement. If $c_{\mathrm{teo}} \neq c_{\mathrm{emp}}$, it would indicate either an error in the CS formula or a violation of RH. The agreement to 99% is strong evidence for both.

Threshold dependence. The lack of a plateau in $c(R_3^{\min})$ is a limitation. A denser grid with higher L values ($L = 30$–80, step 0.1) would reduce d/L^3 contamination and potentially reveal a stable range.

Connection to companion papers. The coefficient $c = 1.245$ was measured empirically in [2], where the physical mechanism (narrowing of the spacing distribution) was also identified. In a separate companion paper [1], we prove that if the convergence pattern holds exactly, the Riemann Hypothesis follows. The present paper provides the first independent theoretical verification of c.

10 Conclusion

We have computed the Berry-Keating correction coefficient c ab initio from the Conrey-Snaith triple correlation formula, obtaining $c = 1.23$ (99% of the empirical value $c = 1.245 \pm 0.040$). This is the first derivation of c from first principles and validates the Berry-Keating conjecture quantitatively. The calculation required 436 evaluations of R_3^{CS} at five values of L, Richardson extrapolation to eliminate the leading a/L correction, and integration against the exact p_2^{GUE} joint density via Schur complement.

The 29 failed perturbative approaches document a fundamental obstruction ($\mathrm{sinc}(1) = 0$) that prevents any kernel-based derivation of c. The Conrey-Snaith route bypasses this by working at the level of the triple correlation function.

Acknowledgements

The author acknowledges the use of publicly available datasets of zeros of the Riemann zeta function: high-precision tables computed by D. J. Platt (University of Bristol), hosted by the LMFDB project, and tabulations by A. M. Odlyzko (University of Minnesota).

The author acknowledges the use of Anthropic's Claude Opus (model claude-opus-4-6) as an AI research assistant throughout this work. Claude was used for data analysis, numerical computation, code development, generation and evaluation of research ideas, identification of mathematical approaches, literature review, and assistance in manuscript preparation. All scientific results, interpretations, and conclusions were independently validated by the author.

Data and code availability

All data, analysis code, and computational scripts are available at `https://github.com/dalarconrub/berry-keating-paper-B`.

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